

# MDGA — Make Durotar Great Again

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Complete documentation for the MDGA guild website: installation, configuration, features, and administration.

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## Recent Updates (v1.2.0)

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- **Guild ↔ Discord Reconciliation:** New admin tab cross-references in-game guild roster against Discord membership. Officers can link mismatched rows to site users, mark rows as "ignored", and paste addon JSON to backfill officer/public notes when the live `/api/addon/sync` endpoint isn't wired up. Backed by `server/routes/reconciliation.js` and `db/migration-041-reconciliation.sql`.
  - **Roster Note Column:** Officer Note column now displayed on the guild roster view. Notes come from the WoW addon (officer/public notes captured client-side and shipped via sync or paste).
  - **Generate Report Countdown:** Member Activity / Guild Gaps report builders now show a robust countdown while the server collates results (replaces the previous spinner-only state).
  - **Addon Schema v4:** Server now accepts both schema v3 and v4. v4 adds `officerNote`, `publicNote`, and `lastSeen` per roster entry.
  - **Addon Allowlist:** Specific characters can be whitelisted ( `ns.ALLOWLIST` in `wow_addon/MDGA/Core.lua` ) to use the addon regardless of in-game rank — useful for testing and for officers whose main is on an alt account.
  - **Officer-check de-duplication:** Addon no longer re-spams the "not an officer" warning on every zone change (tracks `ns.lastOfficerWarnRank` ).
  - **Roster CSV filename:** Guild roster CSV exports now include guild name + timestamp for easier archival.
  - **Guild Verification:** Character validation enforces guild membership in "MAKE DUROTAR GREAT AGAIN" (case-insensitive matching). Duplicate character claims are blocked via a unique `(user_id, character_name, realm_slug)` constraint.
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## 1. Overview

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MDGA is a full-stack web application for the World of Warcraft guild **Make Durotar Great Again** on Tichondrius-US. It provides guild management, a forum, event scheduling, PvP/PvE leaderboards, character profiles synced from Blizzard's API, Discord integration, and a full admin panel.

| Layer            | Technology                                |
|------------------|-------------------------------------------|
| Backend          | Node.js 18+ / Express 4                   |
| Database         | MySQL 8 (mysql2/promise connection pool)  |
| Frontend         | React 19 + React Router 6 (Vite 6 build)  |
| Auth             | Discord OAuth 2.0 + JWT (jsonwebtoken)    |
| Process Manager  | PM2 (auto-restart, logging, systemd boot) |
| External APIs    | Blizzard Game Data API, Discord Bot API   |
| Image Processing | sharp (WebP conversion, resize to 1920px) |
| CSS              | CSS Modules (React components)            |

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## 2. Prerequisites

| Software | Version | Purpose                                 |
|----------|---------|-----------------------------------------|
| Node.js  | 18+     | Runtime                                 |
| npm      | 9+      | Package management                      |
| MySQL    | 8+      | Database                                |
| PM2      | Latest  | Process management (installed by setup) |
| Apache   | 2.4+    | Reverse proxy (production)              |

You will also need accounts for:

- **Discord Developer Portal** — Create an application with OAuth2 + Bot
- **Blizzard Developer Portal** (develop.battle.net) — API credentials for character data
- **Google reCAPTCHA v2** — For the application form

## 3. Installation

### Automated Setup

```
# 1. Clone or upload the project
cd /path/to/MDGA

# 2. Copy and fill in your .env (see Section 4)
cp .env.example .env
nano .env

# 3. Run the setup script
bash setup.sh
```

The setup script handles all 10 steps:

1. Checks prerequisites (Node 18+, npm, MySQL client)
2. Validates `.env` configuration
3. Installs server dependencies ( `npm install` — includes `sharp` for image compression)
4. Installs client dependencies and builds the React frontend
5. Creates `uploads/` and `logs/` directories
6. Initializes the database (schema + all migrations)
7. Seeds a default admin account ( `admin` / `admin` — change immediately)
8. Installs PM2 globally
9. Displays PM2 startup commands
10. Verifies all required environment variables

## Manual Setup

If you prefer to run each step manually:

```
# Install server deps
npm install

# Install client deps and build
cd client && npm install && npm run build && cd ..

# Create directories
mkdir -p uploads logs

# Run database schema
mysql -u USER -p DB_NAME < db/schema.sql

# Run all migrations (in order)
for f in db/migration-*.sql; do mysql -u USER -p DB_NAME < "$f"; done

# Install PM2
npm install -g pm2
```

## 4. Configuration (.env)

Create a `.env` file in the project root. Every variable is documented below.

### Database

| Variable                 | Required | Example                | Description    |
|--------------------------|----------|------------------------|----------------|
| <code>DB_HOST</code>     | Yes      | <code>localhost</code> | MySQL host     |
| <code>DB_PORT</code>     | Yes      | <code>3306</code>      | MySQL port     |
| <code>DB_USER</code>     | Yes      | <code>mdga_user</code> | MySQL username |
| <code>DB_PASSWORD</code> | Yes      | <code>secret</code>    | MySQL password |
| <code>DB_NAME</code>     | Yes      | <code>mdga_core</code> | Database name  |

### Authentication

| Variable                | Required | Example            | Description                                                                       |
|-------------------------|----------|--------------------|-----------------------------------------------------------------------------------|
| <code>JWT_SECRET</code> | Yes      | (64+ random chars) | Secret for signing JWT tokens. Generate with <code>openssl rand -base64 48</code> |

## Discord

| Variable                                | Required | Example                                                 | Description                                           |
|-----------------------------------------|----------|---------------------------------------------------------|-------------------------------------------------------|
| <code>DISCORD_CLIENT_ID</code>          | Yes      | <code>123456789</code>                                  | Discord app Client ID (OAuth2)                        |
| <code>DISCORD_CLIENT_SECRET</code>      | Yes      | <code>abc123 ...</code>                                 | Discord app Client Secret                             |
| <code>DISCORD_BOT_TOKEN</code>          | Yes      | <code>MTQ2 ...</code>                                   | Discord bot token (Bot tab)                           |
| <code>DISCORD_GUILD_ID</code>           | Yes      | <code>123456789</code>                                  | Your Discord server ID (right-click server → Copy ID) |
| <code>DISCORD_OFFICER_CHANNEL_ID</code> | Yes      | <code>123456789</code>                                  | Channel for officer alerts and approval embeds        |
| <code>DISCORD_REDIRECT_URI</code>       | Yes      | <code>https://mdga.dev/api/auth/discord/callback</code> | OAuth callback URL. Must match Discord app settings.  |
| <code>DISCORD_WEBHOOK_URL</code>        | No       | <code>https://discord.com/api/webhooks/ ...</code>      | Webhook for application notifications                 |

## Blizzard API

| Variable                            | Required | Example                  | Description                  |
|-------------------------------------|----------|--------------------------|------------------------------|
| <code>BLIZZARD_CLIENT_ID</code>     | Yes      | <code>105c4e8 ...</code> | Battle.net app Client ID     |
| <code>BLIZZARD_CLIENT_SECRET</code> | Yes      | <code>1UUne1i ...</code> | Battle.net app Client Secret |

## reCAPTCHA

| Variable                          | Required | Example               | Description                                         |
|-----------------------------------|----------|-----------------------|-----------------------------------------------------|
| <code>RECAPTCHA_SITE_KEY</code>   | No       | <code>6Lcv ...</code> | Google reCAPTCHA v2 site key (for application form) |
| <code>RECAPTCHA_SECRET_KEY</code> | No       | <code>6Lcv ...</code> | Google reCAPTCHA v2 secret key                      |

## Email (SMTP)

| Variable               | Required | Example                                    | Description                   |
|------------------------|----------|--------------------------------------------|-------------------------------|
| <code>SMTP_HOST</code> | No       | <code>smtp.gmail.com</code>                | SMTP server                   |
| <code>SMTP_PORT</code> | No       | <code>587</code>                           | SMTP port                     |
| <code>SMTP_USER</code> | No       | <code>you@gmail.com</code>                 | SMTP username                 |
| <code>SMTP_PASS</code> | No       | <code>app-password</code>                  | SMTP password or app password |
| <code>SMTP_FROM</code> | No       | <code>MDGA &lt;noreply@mdga.dev&gt;</code> | From address for emails       |

## Game Configuration

| Variable                    | Required | Example                                          | Description                                                                                         |
|-----------------------------|----------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <code>ALLOWED_REALMS</code> | No       | <code>Tichondrius,Area 52,Illidan,Zul'jin</code> | Comma-separated list of WoW realms users can add characters from. If empty, all realms are allowed. |

## Server

| Variable                 | Required | Example                       | Description                                            |
|--------------------------|----------|-------------------------------|--------------------------------------------------------|
| <code>HOST</code>        | No       | <code>0.0.0.0</code>          | Bind address (default <code>0.0.0.0</code> )           |
| <code>PORT</code>        | No       | <code>3001</code>             | Server port (default <code>3000</code> )               |
| <code>CORS_ORIGIN</code> | Yes      | <code>https://mdga.dev</code> | Allowed CORS origin. Set to your domain in production. |

## Example .env

```
DB_HOST=localhost
DB_PORT=3306
DB_USER=mdga_user
DB_PASSWORD=your_db_password
DB_NAME=mdga_core

JWT_SECRET=gfN0HSArPCyD12xh0SmWzAtndFeE53uZk7JH05GLDy9FnB0YPP ...

DISCORD_CLIENT_ID=123456789
DISCORD_CLIENT_SECRET=your_discord_secret
DISCORD_BOT_TOKEN=MTQ2 ...
DISCORD_GUILD_ID=123456789
DISCORD_OFFICER_CHANNEL_ID=123456789
DISCORD_REDIRECT_URI=https://mdga.dev/api/auth/discord/callback
DISCORD_WEBHOOK_URL=https://discord.com/api/webhooks/ ...

BLIZZARD_CLIENT_ID=your_blizzard_id
BLIZZARD_CLIENT_SECRET=your_blizzard_secret

ALLOWED_REALMS=Tichondrius,Area 52,Illidan,Zul'jin

RECAPTCHA_SITE_KEY=6Lcv ...
RECAPTCHA_SECRET_KEY=6Lcv ...

SMTP_HOST=smtp.gmail.com
SMTP_PORT=587
SMTP_USER=you@gmail.com
SMTP_PASS=your_app_password
SMTP_FROM=MDGA <noreply@mdga.dev>

HOST=0.0.0.0
PORT=3001
CORS_ORIGIN=https://mdga.dev
```

---

## 5. Running the Server

### Production (PM2)

PM2 is the recommended way to run in production. It auto-restarts on crash, manages logs, and survives server reboots.

```
# Start the application
pm2 start ecosystem.config.js

# Enable auto-start on server boot
pm2 startup systemd
# (Run the sudo command it prints)
pm2 save

# Useful PM2 commands
pm2 status          # View running processes
pm2 logs mdga       # Tail application logs
pm2 logs mdga --lines 100 # Last 100 lines
pm2 restart mdga     # Restart the app
pm2 stop mdga        # Stop the app
pm2 monit            # Real-time monitoring dashboard
```

PM2 configuration is in `ecosystem.config.js`:

- Auto-restart with exponential backoff (1s → 2s → 4s... up to 15s)
- 512MB memory limit (restarts if exceeded)
- Logs written to `logs/pm2-out.log` and `logs/pm2-error.log`
- `interpreter` pinned to the nvm Node 24 binary so pm2 never falls back to the system `/usr/bin/node` (which is Node 12 on the current production box and cannot parse modern syntax). Override with `PM2_INTERPRETER=/path/to/node` if the nvm install moves.

## Development

```
# Backend with auto-restart on file changes
npm run dev

# Frontend dev server with hot module replacement (separate terminal)
cd client && npm run dev
```

The Vite dev server runs on port 5173 and proxies `/api`, `/uploads`, and `/images` to the backend on port 3001.

## Rebuilding After Frontend Changes

```
cd client && npm run build
```

No server restart needed — Express serves files directly from `client/dist/`.



## 6. Apache Reverse Proxy

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In production, Apache sits in front of Node.js. The `.htaccess` file handles routing:

```
# API and file requests → Node.js on port 3001
RewriteRule ^api/(.*)$ http://localhost:3001/api/$1 [P,L]
RewriteRule ^uploads/(.*)$ http://localhost:3001/uploads/$1 [P,L]
RewriteRule ^images/(.*)$ http://localhost:3001/images/$1 [P,L]

# Everything else → React SPA (index.html)
RewriteRule ^ /index.html [L]
```

Required Apache modules:

```
sudo a2enmod proxy proxy_http rewrite
sudo systemctl restart apache2
```

If using a VirtualHost config instead of `.htaccess`, include `ProxyPreserveHost On` and match the same rules.

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## 7. Architecture

### Directory Structure

```

MDGA/
├─ server/
│   ├─ index.js           # Express entry point, middleware, static serving
│   ├─ db.js             # MySQL connection pool
│   ├─ blizzard.js       # Blizzard API client (OAuth, character data)
│   ├─ bot.js            # Discord bot (approval flow, alerts, nickname sync)
│   ├─ ipban.js          # IP ban middleware
│   ├─ middleware/
│   │   ├─ auth.js       # JWT auth, role/permission middleware
│   │   └─ upload.js     # Multer + magic byte validation + sharp compression
│   └─ routes/
│       ├─ auth.js       # Login, logout, password, /me
│       ├─ discord.js    # Discord OAuth2 flow + role sync
│       ├─ discord-roles.js # Discord-to-site role mapping
│       ├─ config.js     # Public config (allowed realms)
│       ├─ events.js     # Event CRUD + RSVP
│       ├─ applications.js # Guild applications
│       ├─ forum.js      # Forum categories, posts, comments, votes
│       ├─ users.js      # User management (admin)
│       ├─ characters.js  # Character CRUD + Blizzard validation
│       ├─ profile.js    # User profile endpoint
│       ├─ leaderboard.js # PvP/PvE leaderboard + stat refresh
│       ├─ roles.js      # RBAC role management
│       ├─ dashboard.js  # Dashboard data
│       ├─ guild.js      # Guild roster + member management
│       ├─ carousel.js   # Carousel images + site settings
│       ├─ upload.js     # Image upload endpoint
│       ├─ reports.js    # Admin reports + analytics
│       ├─ reconciliation.js # Guild ↔ Discord reconciliation + addon paste ingest
│       └─ addon.js      # WoW addon sync endpoint
│   └─ services/
│       ├─ character-sync.js # Character stat refresh scheduler
│       ├─ guild-sync.js    # Guild roster sync from Blizzard
│       └─ email.js        # SMTP email sending
├─ client/                # React SPA (Vite)
│   └─ src/
│       ├─ App.jsx        # Router definition
│       └─ contexts/
│           └─ AuthContext.jsx # Auth state, API fetch, permissions
│       └─ components/
│           ├─ layout/    # Layout, Nav, Footer
│           └─ common/    # ProtectedRoute, PageHero, LoadingSpinner

```

```

| | | └─ pages/
| | |   └─ Home/      # Dashboard + hero
| | |   └─ Login/     # Discord OAuth login
| | |   └─ AdminLogin/ # Password login (admin fallback)
| | |   └─ Join/      # Application form
| | |   └─ Story/     # Guild lore timeline
| | |   └─ Leadership/ # Leadership roster
| | |   └─ Events/    # Calendar + RSVP
| | |   └─ Forum/     # Index, Category, Post, NewPost
| | |   └─ Leaderboards/ # PvP/PvE rankings
| | |   └─ Profile/   # User profile + characters
| | |   └─ Admin/     # Admin panel (6+ tabs)
| | └─ hooks/        # useDocumentTitle
| | └─ utils/        # helpers, timezone
| | └─ data/         # leaderboardData, wowData
| └─ vite.config.js  # Build config + dev proxy
└─ db/              # SQL schema + 40 migration files
└─ uploads/         # User-uploaded images (compressed WebP)
└─ images/          # Static site images
└─ logs/            # PM2 log files
└─ ecosystem.config.js # PM2 process config
└─ setup.sh         # Automated setup script
└─ .htaccess        # Apache proxy rules
└─ docker-compose.yml # MySQL container (optional)
└─ .env             # Environment variables (never commit)

```

## How Serving Works

The Express server serves the React SPA from `client/dist/` when the build exists. All `/api/*` routes are handled by Express. Non-API requests fall through to `index.html` for React Router to handle client-side routing.

In production, Apache proxies `/api/*`, `/uploads/*`, and `/images/*` to Node.js on port 3001, and serves everything else through the SPA fallback.

## 8. Authentication & Login

### Discord OAuth 2.0 (Primary)

This is the main login method for all users.

#### Flow:

1. User clicks **Login with Discord** on `/login`
2. Redirects to `/api/auth/discord?from=login`
3. Server generates a CSRF state token (10-min TTL) and redirects to Discord

4. User authorizes the app on Discord
5. Discord redirects back to `/api/auth/discord/callback`
6. Server validates state, exchanges code for Discord access token
7. Fetches Discord profile ( `identify email` scopes)
8. Finds or creates user by `discord_id`
9. Checks guild membership via Discord bot:
  - **In guild** → status = `active` , syncs rank from `discord_role_mappings`
  - **Not in guild** → status = `pending_approval` , sends approval embed to officer channel
10. Creates a one-time login grant (2-min TTL)
11. Redirects to `/login/#code=<grant>` (hash fragment to prevent Referer leakage)
12. Frontend exchanges grant for JWT + user data via `POST /api/auth/discord/exchange`
13. Token and user stored in `sessionStorage`

## Username/Password Login (Admin Fallback)

Available at `/admin-login` for emergencies when Discord is down. Uses `POST /api/auth/login` .

Protected by:

- 5 failed attempts per IP → 15-minute logout
- 60 failed attempts in 24h → 24-hour site-wide IP ban
- Discord alerts sent to officer channel on lockouts and bans

## JWT Token

```
{
  id: 42,
  username: "Thrall",
  rank: "officer",      // recruit | member | veteran | officer | guildmaster
  permissions: ["forum.create_posts", "admin.view_panel", ...]
}
// Expires in 24 hours
```

## Changing Passwords

Logged-in users can set or change their password via `PUT /api/auth/password` . Requires the current password (if one exists) and a new password of 8+ characters. Passwords are hashed with bcrypt (12 rounds).

---

# 9. Discord Integration

## Bot Features

The Discord bot ( `server/bot.js` ) provides:

| Feature                     | Description                                                                      |
|-----------------------------|----------------------------------------------------------------------------------|
| <b>Approval flow</b>        | Sends embed with Approve/Reject buttons to officer channel for non-guild members |
| <b>Nickname sync</b>        | Sets Discord nickname to match main WoW character name                           |
| <b>Kick/leave detection</b> | Suspends user account when member leaves Discord guild                           |
| <b>Officer alerts</b>       | Sends alerts for login lockouts, IP bans, rank changes, and security events      |
| <b>Role fetching</b>        | Provides live Discord roles list for the role mapping UI                         |
| <b>Invite links</b>         | Creates one-time Discord invite links (24h, 1 use) for approved members          |

## Discord Role Sync

On each login, the server:

1. Fetches the user's Discord roles from the guild
2. Looks up `discord_role_mappings` for matches
3. Determines highest-priority rank: recruit → member → veteran → officer → guildmaster
4. Updates the user's `rank` column
5. Syncs RBAC roles (adds matched roles, removes unmatched synced roles, leaves manual roles)

## Setting Up the Discord Bot

1. Go to [Discord Developer Portal](#)
2. Create a new application
3. **OAuth2** tab: Add redirect URI matching your `DISCORD_REDIRECT_URI`
4. **Bot** tab: Create a bot, copy the token for `DISCORD_BOT_TOKEN`
5. **Bot** tab: Enable "Server Members Intent" and "Message Content Intent"
6. Invite the bot to your server with `bot` + `applications.commands` scopes and these permissions:
  - Manage Nicknames
  - Create Instant Invite
  - Send Messages
  - Embed Links
  - Read Message History

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## 10. Roles & Permissions (RBAC)

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### Guild Ranks

Legacy rank system, synced from Discord. Determines display badges and basic access levels.

| Rank        | Level | Description                   |
|-------------|-------|-------------------------------|
| recruit     | 0     | New member, pending promotion |
| member      | 1     | Regular guild member          |
| veteran     | 2     | Experienced member            |
| officer     | 3     | Guild leadership              |
| guildmaster | 4     | Full access to everything     |

## RBAC Roles

Fine-grained permission system layered on top of ranks. Users can have multiple roles. Permissions are additive. Guildmaster bypasses all permission checks.

| Permission                      | Category    | Description                    |
|---------------------------------|-------------|--------------------------------|
| forum.create_posts              | Forum       | Create forum posts             |
| forum.create_comments           | Forum       | Reply to posts                 |
| forum.vote                      | Forum       | Vote on posts/comments         |
| forum.pin_posts                 | Forum       | Pin/unpin posts                |
| forum.lock_posts                | Forum       | Lock/unlock posts              |
| forum.delete_any_post           | Forum       | Delete any post                |
| forum.delete_any_comment        | Forum       | Delete any comment             |
| forum.access_officer_categories | Forum       | Access officer-only categories |
| forum.manage_categories         | Forum       | Create/edit categories         |
| events.manage                   | Events      | Create/edit/delete events      |
| admin.view_panel                | Admin       | View admin panel               |
| admin.manage_applications       | Admin       | Review guild applications      |
| admin.manage_users              | Admin       | Change ranks, ban/delete users |
| admin.manage_roles              | Admin       | Create/edit/delete roles       |
| leaderboard.bulk_refresh        | Leaderboard | Refresh all character stats    |
| guild.manage                    | Guild       | Manage guild sync and roster   |
| guild.view_roster               | Guild       | View full guild roster         |

## Default Roles

| Role          | Default             | Key Permissions                                              |
|---------------|---------------------|--------------------------------------------------------------|
| Member        | Yes (auto-assigned) | Create posts, comments, vote, view roster                    |
| Moderator     | No                  | Pin, lock, delete any post/comment, officer categories       |
| Event Manager | No                  | Create/manage events                                         |
| Officer       | No                  | All moderator + admin panel, manage apps/users, bulk refresh |
| Guild Master  | No                  | All permissions                                              |

## Auth Middleware

```
requireAuth           // JWT required
requireOfficer        // Rank officer/GM OR admin.view_panel permission
requireGuildMaster    // Rank guildmaster only
requirePermission('perm.key') // Check specific permission
```

# 11. Characters & Blizzard API

## Adding a Character

1. Navigate to your **Profile** page
2. Click **Add Character**
3. Select a realm from the dropdown (configured via `ALLOWED_REALMS` in `.env`)
4. Enter the character name and click **Search**
5. The server validates the character against Blizzard's Armory:
  - Must exist on a supported realm
  - Must be a current member of **Make Durotar Great Again**
6. If validated, review the character details and click **Save Character**
7. Optionally check "Set as main character" to make it your primary

## Validation Rules

All character additions are validated server-side against the Blizzard Armory:

| Check                   | Error Message                                                                                                                                                   |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Unsupported realm       | "This realm is not supported. Characters must be on one of the following realms: ..."                                                                           |
| Character not found     | "Character not found on World of Warcraft Armory"                                                                                                               |
| Wrong guild             | "This character is in <Other Guild>, not <Make Durotar Great Again>. Only members of our guild can be added."                                                   |
| No guild                | "This character is not in a guild. Only members of <Make Durotar Great Again> can be added."                                                                    |
| Already on your profile | "This character is already on your profile."                                                                                                                    |
| Claimed by another user | "This character is already claimed by {discord_username}. If this is your character, please open a support ticket in our Discord server to have it reassigned." |

**Security Note:** Guild membership is validated on every character addition (both lookup and save endpoints). Duplicate character claims are prevented with a unique constraint on `(user_id, character_name, realm_slug)`.

## Realm Configuration

Allowed realms are set in `.env`:

```
ALLOWED_REALMS=Tichondrius,Area 52,Illdan,Zul'jin
```

To add more realms, append them to the comma-separated list and restart the server. The frontend automatically fetches the allowed list from `GET /api/config/realms`.

## Blizzard API Data

The server fetches character data from the Blizzard Game Data API using Client Credentials OAuth.

### Per-character data fetched:

| Category         | Fields                                                                     |
|------------------|----------------------------------------------------------------------------|
| Profile          | Name, level, race, class, spec, item level, achievement points, last login |
| Media            | Character render image URL                                                 |
| PvP Summary      | Solo Shuffle, 2v2, 3v3, RBG ratings                                        |
| PvP Brackets     | Individual bracket ratings and W/L records                                 |
| Statistics       | Honorable kills, killing blows, arenas played/won, BGs played/won, deaths  |
| PvE              | Dungeons entered, raids entered, quests completed, creatures killed        |
| Mythic+          | M+ rating, highest key level completed                                     |
| Raid Progression | Latest raid mythic boss kills                                              |
| Talents          | Active spec, loadout name, talent selections                               |



## Auto-Refresh

Character stats refresh automatically:

- Every 2 hours after server boot
  - Staggered 5 minutes between characters to avoid API throttling
  - Users can manually refresh via the **Refresh Stats** button on their profile
  - Officers can trigger a guild-wide refresh from the admin panel
- 

## 12. Guild System

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### Guild Sync

The server periodically syncs the full guild roster from Blizzard:

- Fetches all guild members (name, realm, level, class, race, rank)
- Cross-links guild members to site users by matching character name + realm
- Stores data in the `guild_members` table
- Syncs guild achievements to `guild_achievements`

### Guild Roster (Admin)

Officers can view the full guild roster at **Admin > Guild**:

- Search by character name
- Filter by class, rank, level
- View linked site accounts
- Ban/unban guild members

### Guild Activity Feed

Tracks real-time guild events:

- Rank changes
  - Member joins/leaves
  - Online/offline status (via addon)
- 

## 13. Leaderboards

---

The `/leaderboards` page displays guild-wide rankings across multiple brackets.

### PvP Brackets

- Solo Shuffle, Arena 3v3, Arena 2v2, RBG Rating

- Honorable Kills, Killing Blows
- Arenas Played/Won, Battlegrounds Played/Won

## PvE Brackets

- Mythic+ Rating, Item Level
- Highest M+ Key, Mythic Bosses Killed
- Dungeons Entered, Raids Entered, Creatures Killed

## General Brackets

- Total Deaths, Quests Completed, Achievement Points

## Expandable Details

Click any row in the leaderboard to expand and view the character's full stat card with PvP, PvE, and General statistics in a 3-column grid layout.

## Refreshing Stats

- Members can refresh their own stats (15-minute cooldown)
- Officers with `leaderboard.bulk_refresh` permission can trigger a guild-wide refresh

---

# 14. Forum

## Structure

- **Categories** — Officer-created, with optional officer-only access
- **Posts** — Support text content and one image attachment
- **Comments** — Replies to posts, also support images
- **Votes** — Upvote/downvote on both posts and comments

## Features

| Feature                   | Description                                                             |
|---------------------------|-------------------------------------------------------------------------|
| <b>Sorting</b>            | Hot (Reddit-style), New, Top                                            |
| <b>Pagination</b>         | Configurable page size (10/15/20/50)                                    |
| <b>Search</b>             | Full-text search across posts, comments, and usernames (300ms debounce) |
| <b>View tracking</b>      | Unique views per user per post                                          |
| <b>Pinned posts</b>       | Stick important posts to the top                                        |
| <b>Locked posts</b>       | Prevent new comments                                                    |
| <b>Officer categories</b> | Restrict access to officers+                                            |
| <b>Reporting</b>          | Members can report posts/comments for review                            |
| <b>Image attachments</b>  | One image per post or comment (auto-compressed to WebP)                 |

## Creating a Post

1. Navigate to `/forum`
2. Click a category
3. Click **New Post**
4. Enter a title (max 200 chars), content, and optional image
5. Submit

## Moderation

Officers/moderators can:

- **Pin** posts to keep them at the top of a category
- **Lock** posts to prevent further comments
- **Delete** any post or comment
- **Review reports** in the admin panel

---

## 15. Events

### Viewing Events

The `/events` page is public — anyone can see upcoming guild events.

### Creating Events

Users with `events.manage` permission can create events:

| Field          | Description                                                     |
|----------------|-----------------------------------------------------------------|
| Title          | Event name (max 150 chars)                                      |
| Description    | Full event details                                              |
| Category       | PvP, Defense, Social, or Raid                                   |
| Start/End Time | Date and time with timezone support                             |
| Recurring      | Optional: weekly, biweekly, or custom interval (2-52 instances) |

## RSVP

Logged-in members can RSVP with three statuses:

- **Going** — Confirmed attendance
- **Maybe** — Tentative
- **Not Going** — Declined

RSVP counts are displayed on event cards. Officers can view the full attendee list.

## Managing Events

- Edit or delete individual events
- Delete an entire recurring series (future events only)

---

# 16. Applications (Recruitment)

---

## How It Works

1. Prospective member visits `/join`
2. Fills out the application form:
  - Character name, server, class/spec
  - Discord tag
  - WoW experience
  - Why they want to join
  - reCAPTCHA verification
3. Application is submitted (no login required)
4. Notification sent to Discord webhook
5. Officers review in **Admin > Applications**
6. Officer approves or denies:
  - **Approve**: User account activated, approval email sent with Discord invite link
  - **Deny**: Application marked as denied

## Discord Approval Flow

If a user logs in with Discord but isn't in the guild:

1. Account created with `pending_approval` status
2. Embed with Approve/Reject buttons sent to officer channel
3. Officer clicks button → account activated or rejected

---

## 17. Profile System

### User Profile ( `/profile` )

Every member has a profile page showing:

| Section     | Content                                                                 |
|-------------|-------------------------------------------------------------------------|
| Hero        | Avatar, display name, rank badge, Discord username, join date, timezone |
| Forum Stats | Total posts, views, and comments                                        |
| Characters  | Card grid of all WoW characters                                         |

### Character Cards

Each character card shows:

- **Front:** Character render, name, realm, guild, class/spec, level, item level, quick PvP pills (Shuffle, 3v3, 2v2, RBG, HKs), Armory link
- **Back** (click to flip): Full stat card — all PvP ratings, arena/BG records with win rates, PvE stats, achievement points, last synced time

### Profile Actions (Own Profile)

- **Add Character** — Search and validate from Blizzard
- **Set Main** — Designate primary character (syncs to Discord nickname)
- **Delete Character** — Remove a character from your profile
- **Refresh Stats** — Re-fetch all character stats from Blizzard
- **Change Timezone** — Click your timezone to change it

---

## 18. Admin Panel

Accessible at `/admin` for officers and users with `admin.view_panel` permission.

## Tab: Members

- View all registered users with rank badges
- Search by username or display name
- **Change Rank** — Promote/demote (hierarchy enforced — can't promote above your own rank)
- **Ban** — Ban with reason (sends Discord alert)
- **Delete** — Remove user account
- **Resend Invite** — Re-send approval email with Discord invite

## Tab: Applications

- View pending, approved, and denied applications
- **Approve** — Activates user, sends approval email with Discord invite
- **Deny** — Marks application as denied

## Tab: Forum Management

- **Create Category** — Add new forum categories (name, description, officer-only toggle, sort order)
- **Reports** — Review reported posts/comments (open, reviewing, resolved, dismissed)

## Tab: Events

- Create, edit, and delete events
- Manage recurring event series

## Tab: Carousel

- **Upload Images** — Add screenshots/photos to the homepage carousel
- **Manage Order** — Drag to reorder, edit alt text
- **Home Background** — Upload or set URL for the hero background image

## Tab: Reports

- **Member Activity Report** — Search and filter users by rank, status, date range, activity period. Export results.
- **Guild Gaps Report** — Compare Discord members vs. site accounts. Identify: needs Discord link, no site account, Discord not active, etc.
- **Generate Report** — Long-running reports show a live countdown timer while collating server-side; results stream into the table when ready.
- **Saved Presets** — Save and load report filter configurations

## Tab: Reconciliation

- Cross-references the in-game guild roster (from Blizzard + addon) against Discord membership and site accounts

- Each row flags its state: linked, needs link, no site account, Discord-only, etc.
- **Link** — Manually link a guild member row to an existing site user
- **Ignore / Un-ignore** — Mark a row as intentionally unlinked (alt characters, etc.)
- **Refresh** — Re-runs `discord-member-sync` + `guild-sync` services
- **Paste Addon JSON** — Officers can paste the addon's "Export JSON" output to backfill officer notes, public notes, and `LastSeen` timestamps onto `guild_members` rows
- **Officer Note column** — Surfaces notes captured client-side by the addon

### Tab: Roles (Guild Master Only)

- Create/edit/delete custom RBAC roles
- Toggle permissions per role
- Assign roles to specific users

### Tab: Discord Roles (Guild Master Only)

- View all Discord server roles (live from bot)
- Map Discord roles → site ranks (recruit/member/veteran/officer/guildmaster)
- Map Discord roles → RBAC roles

---

## 19. Home Page & Carousel

---

### Visitor View

Non-logged-in visitors see:

- Hero section with CTAs to join or read the guild story
- Stats bar, guild photo, about section
- Gallery carousel
- Leadership preview
- PvP achievements highlight

### Member Dashboard

Logged-in members see a personalized dashboard:

- Welcome message with rank badge
- Quick stats (forum posts, replies, character count, top PvP rating)
- Recent forum activity (last 10 posts)
- Upcoming events with RSVP counts
- Latest Updates feed with post previews

## Carousel Management

Officers manage the carousel from **Admin > Carousel**:

- Upload images (up to 15MB raw — auto-compressed to WebP)
  - Or provide an external URL
  - Set alt text for accessibility
  - Control sort order
  - Auto-advances every 5 seconds with prev/next navigation
- 

## 20. Image Uploads

---

### How It Works

All image uploads go through the same pipeline:

1. **Multer** accepts the file (up to 15MB raw)
2. **MIME type check** — Only JPEG, PNG, GIF, and WebP allowed
3. **Magic byte detection** — Validates actual file content matches declared type
4. **sharp compression** — Resizes to max 1920px wide and converts to WebP at 80% quality
5. **Saved** with a timestamp + random hex filename to `uploads/`

GIF files are stored as-is (to preserve animations).

### Where Uploads Are Used

- Forum posts and comments (image attachments)
- Carousel images (homepage gallery)
- Home background image

### Storage

Uploaded files are saved to the `uploads/` directory and served via `/uploads/` URL path. Files are served with:

- `X-Content-Type-Options: nosniff`
  - `Cache-Control: public, max-age=86400` (24-hour browser cache)
- 

## 21. WoW Addon Integration

---

The project includes a companion WoW addon that sends real-time guild data to the website.



## What the Addon Tracks

| Event Type               | Description                   |
|--------------------------|-------------------------------|
| <code>rank_change</code> | Guild member promoted/demoted |
| <code>join</code>        | New member joined the guild   |
| <code>leave</code>       | Member left or was kicked     |
| <code>online</code>      | Member logged in              |
| <code>offline</code>     | Member logged off             |

## Sync Endpoint

`POST /api/addon/sync` — Officer-only (guild rank 0-2), rate-limited to 10 requests per 15 minutes.

## Validation

- Schema version 3 or 4 accepted (v4 adds `officerNote`, `publicNote`, `lastSeen` per roster entry)
- Guild name must match (case-insensitive)
- Data must be fresh (30min for capture, 2h for events)
- Character ownership verified
- Max 100 events and 1000 roster entries per sync

## Officer Gate & Allowlist

The addon only activates if the player is an officer (in-game guild rank  $\leq$  `OFFICER_RANK_THRESHOLD`, default 2). Specific characters can be allowlisted to bypass the rank check via `ns.ALLOWLIST` in

`wow_addon/MDGA/Core.lua` — matched case-insensitively against the player name (server suffix ignored). The "not an officer" warning is de-duplicated per rank to avoid spam on zone changes.

## Paste-Based Ingest (Fallback)

For officers without the sync endpoint wired up, the addon exposes an "Export JSON" button. Officers can paste the output into **Admin > Reconciliation** to backfill officer/public notes and `lastSeen` onto `guild_members`. Server endpoint: `POST /api/reconciliation/addon-paste` (deduplicated by SHA-256 of the payload).

---

## 22. Reports & Analytics

### Member Activity Report

Available at **Admin > Reports**. Provides insights into member engagement:

- Search by username/display name

- Filter by rank, account status, date range
- Activity period filter (7 days, 30 days, 90 days, all time)
- Sort by any column
- Export all results
- Save/load filter presets

## Guild Gaps Report

Cross-references Discord members with site accounts:

| State              | Meaning                                  |
|--------------------|------------------------------------------|
| Needs Discord      | Site account has no Discord link         |
| No Site Account    | Discord member hasn't registered on site |
| No Discord Link    | Registered but Discord not linked        |
| Discord Not Active | Linked but no longer in Discord server   |
| Linked Active      | Everything is connected                  |

## Forum Reports

Review reported posts and comments:

- Status: Open → Reviewing → Resolved / Dismissed
  - View reporter, reason, and reported content
- 

## 23. Security

---

### Rate Limiting

| Scope            | Limit             | Window     |
|------------------|-------------------|------------|
| All API routes   | 200 requests      | 15 minutes |
| Login attempts   | 5 failures per IP | 15 minutes |
| IP ban threshold | 60 failures       | 24 hours   |
| Addon sync       | 10 requests       | 15 minutes |

### IP Banning

After 60 failed login attempts from a single IP within 24 hours, the IP is banned site-wide for 24 hours. Bans are stored in memory (reset on server restart). Discord alerts are sent to the officer channel.

## Security Headers (Helmet.js)

- Content Security Policy (restricts scripts, styles, fonts, images, connections)
- `X-Content-Type-Options: nosniff`
- Cross-Origin policies

## Image Upload Security

1. MIME type whitelist (JPEG, PNG, GIF, WebP only)
2. Magic byte detection (validates actual binary content)
3. MIME vs. content type cross-check (prevents extension spoofing)
4. Randomized filenames (timestamp + crypto random hex)
5. Server-side compression (prevents oversized file storage)

## CSRF Protection

Discord OAuth uses random state tokens with 10-minute TTL, stored server-side and validated on callback.

## Database Security

- Parameterized queries throughout (no string concatenation in SQL)
- LIKE wildcards escaped ( `%`, `_`, `\` )
- Foreign keys with CASCADE/SET NULL

---

## 24. Database

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### Schema

All migrations are in `db/` (schema.sql + migration-001 through migration-041). Run automatically by `setup.sh`.

## Core Tables

| Table                              | Purpose                                                     |
|------------------------------------|-------------------------------------------------------------|
| <code>users</code>                 | User accounts (Discord/password auth, rank, status, avatar) |
| <code>user_characters</code>       | Characters linked to users (Blizzard Armory data)           |
| <code>pvp_stats</code>             | Per-character PvP/PvE statistics                            |
| <code>guilds</code>                | Guild data from Blizzard API                                |
| <code>guild_members</code>         | Full guild roster (synced from Blizzard)                    |
| <code>guild_member_stats</code>    | Guild-wide leaderboard stats                                |
| <code>guild_achievements</code>    | Guild achievement data                                      |
| <code>guild_activity</code>        | Join/leave/rank change activity feed                        |
| <code>events</code>                | Calendar events (supports recurring)                        |
| <code>event_rsvps</code>           | RSVP attendance tracking                                    |
| <code>forum_categories</code>      | Forum structure (officer-only flag, sort order)             |
| <code>forum_posts</code>           | Posts (pinned, locked, view count, image)                   |
| <code>forum_comments</code>        | Post replies                                                |
| <code>forum_votes</code>           | Post votes (+1/-1)                                          |
| <code>forum_comment_votes</code>   | Comment votes (+1/-1)                                       |
| <code>forum_post_views</code>      | Unique view tracking per user                               |
| <code>forum_reports</code>         | Reported content (status workflow)                          |
| <code>applications</code>          | Guild applications (pending/approved/denied)                |
| <code>roles</code>                 | RBAC role definitions                                       |
| <code>permissions</code>           | RBAC permission definitions                                 |
| <code>role_permissions</code>      | Role ↔ Permission mapping                                   |
| <code>user_roles</code>            | User ↔ Role assignment                                      |
| <code>discord_role_mappings</code> | Discord role → site rank/RBAC mapping                       |
| <code>login_attempts</code>        | Failed login tracking per IP                                |
| <code>carousel_images</code>       | Homepage carousel images                                    |
| <code>site_settings</code>         | Key/value configuration (home background, etc.)             |
| <code>addon_events</code>          | Real-time guild events from WoW addon                       |
| <code>game_rank_mappings</code>    | In-game guild rank → site rank mapping                      |
| <code>user_report_presets</code>   | Saved admin report filter configurations                    |

| Table                        | Purpose                                                                        |
|------------------------------|--------------------------------------------------------------------------------|
| <code>discord_members</code> | Cached Discord guild member list for reconciliation against the in-game roster |
| <code>addon_ingests</code>   | SHA-256 ledger of addon JSON pastes (deduplicates re-submissions)              |

## Running Migrations Manually

```
# Single migration
mysql -u USER -p DB_NAME < db/migration-040-fix-charset-utf8mb4.sql

# All migrations (skips already-applied)
for f in db/migration-*.sql; do mysql -u USER -p DB_NAME < "$f" 2>/dev/null; done
```

## 25. API Reference

### Authentication

| Method | Endpoint                                       | Auth    | Description                 |
|--------|------------------------------------------------|---------|-----------------------------|
| GET    | <code>/api/auth/discord?from=login join</code> | —       | Start Discord OAuth         |
| GET    | <code>/api/auth/discord/callback</code>        | —       | Discord OAuth callback      |
| POST   | <code>/api/auth/discord/exchange</code>        | —       | Exchange grant code for JWT |
| POST   | <code>/api/auth/login</code>                   | —       | Username/password login     |
| PUT    | <code>/api/auth/password</code>                | JWT     | Change password             |
| POST   | <code>/api/auth/logout</code>                  | JWT     | Logout                      |
| GET    | <code>/api/auth/me</code>                      | JWT     | Current user + permissions  |
| POST   | <code>/api/auth/test-email</code>              | Officer | Test SMTP configuration     |

### Config

| Method | Endpoint                        | Auth | Description                 |
|--------|---------------------------------|------|-----------------------------|
| GET    | <code>/api/config/realms</code> | —    | Get allowed WoW realms list |

### Dashboard

| Method | Endpoint                    | Auth | Description                 |
|--------|-----------------------------|------|-----------------------------|
| GET    | <code>/api/dashboard</code> | JWT  | Personalized home dashboard |

## Characters

| Method | Endpoint                 | Auth | Description                         |
|--------|--------------------------|------|-------------------------------------|
| POST   | /api/characters/lookup   | JWT  | Search character on Blizzard Armory |
| GET    | /api/characters/:userId  | JWT  | Get user's characters               |
| POST   | /api/characters          | JWT  | Add validated character             |
| PUT    | /api/characters/:id      | JWT  | Update character                    |
| PUT    | /api/characters/:id/main | JWT  | Set as main character               |
| DELETE | /api/characters/:id      | JWT  | Delete character                    |

## Profile

| Method | Endpoint              | Auth | Description                                      |
|--------|-----------------------|------|--------------------------------------------------|
| GET    | /api/profile/:id      | JWT  | Full profile (user, characters, stats, activity) |
| PUT    | /api/profile/timezone | JWT  | Update timezone                                  |

## Leaderboard

| Method | Endpoint                                         | Auth                     | Description                 |
|--------|--------------------------------------------------|--------------------------|-----------------------------|
| GET    | /api/leaderboard?<br>bracket=solo_shuffle&page=1 | JWT                      | Rankings for a bracket      |
| POST   | /api/leaderboard/refresh                         | JWT                      | Refresh own character stats |
| POST   | /api/leaderboard/refresh-all                     | leaderboard.bulk_refresh | Guild-wide stat refresh     |

## Forum

| Method | Endpoint                                     | Auth                    | Description                           |
|--------|----------------------------------------------|-------------------------|---------------------------------------|
| GET    | <code>/api/forum/categories</code>           | —                       | List categories                       |
| POST   | <code>/api/forum/categories</code>           | forum.manage_categories | Create category                       |
| GET    | <code>/api/forum/categories/:id/posts</code> | —                       | Posts in category (sort: hot/new/top) |
| GET    | <code>/api/forum/search?q=term</code>        | —                       | Full-text search                      |
| POST   | <code>/api/forum/posts</code>                | JWT                     | Create post                           |
| GET    | <code>/api/forum/posts/:id</code>            | —                       | Get post + comments + votes           |
| POST   | <code>/api/forum/posts/:id/comments</code>   | JWT                     | Add comment                           |
| POST   | <code>/api/forum/posts/:id/vote</code>       | JWT                     | Vote (+1/-1/0)                        |
| POST   | <code>/api/forum/comments/:id/vote</code>    | JWT                     | Vote on comment                       |
| POST   | <code>/api/forum/posts/:id/report</code>     | JWT                     | Report post                           |
| POST   | <code>/api/forum/comments/:id/report</code>  | JWT                     | Report comment                        |
| DELETE | <code>/api/forum/posts/:id</code>            | JWT                     | Delete post                           |
| DELETE | <code>/api/forum/comments/:id</code>         | JWT                     | Delete comment                        |
| PUT    | <code>/api/forum/posts/:id/pin</code>        | forum.pin_posts         | Toggle pin                            |
| PUT    | <code>/api/forum/posts/:id/lock</code>       | forum.lock_posts        | Toggle lock                           |

## Events

| Method | Endpoint                                  | Auth          | Description                    |
|--------|-------------------------------------------|---------------|--------------------------------|
| GET    | <code>/api/events</code>                  | —             | List upcoming events           |
| POST   | <code>/api/events</code>                  | events.manage | Create event                   |
| PUT    | <code>/api/events/:id</code>              | events.manage | Update event                   |
| DELETE | <code>/api/events/:id</code>              | events.manage | Delete event                   |
| DELETE | <code>/api/events/series/:seriesId</code> | events.manage | Delete future events in series |
| POST   | <code>/api/events/:id/rsvp</code>         | JWT           | RSVP to event                  |
| GET    | <code>/api/events/:id/rsvps</code>        | JWT           | Get RSVP list                  |

## Applications

| Method | Endpoint                                      | Auth                      | Description        |
|--------|-----------------------------------------------|---------------------------|--------------------|
| POST   | <code>/api/applications</code>                | —                         | Submit application |
| GET    | <code>/api/applications?status=pending</code> | admin.manage_applications | List applications  |
| PUT    | <code>/api/applications/:id</code>            | admin.manage_applications | Approve or deny    |

## Users

| Method | Endpoint                                  | Auth               | Description         |
|--------|-------------------------------------------|--------------------|---------------------|
| GET    | <code>/api/users</code>                   | admin.manage_users | List all users      |
| GET    | <code>/api/users/banned</code>            | admin.manage_users | List banned users   |
| PUT    | <code>/api/users/:id/rank</code>          | admin.manage_users | Change rank         |
| PUT    | <code>/api/users/:id/ban</code>           | admin.manage_users | Ban user            |
| PUT    | <code>/api/users/:id/unban-request</code> | admin.manage_users | Request unban       |
| DELETE | <code>/api/users/:id</code>               | admin.manage_users | Delete user         |
| POST   | <code>/api/users/:id/resend-invite</code> | admin.manage_users | Resend invite email |

## Guild

| Method | Endpoint                        | Auth              | Description                          |
|--------|---------------------------------|-------------------|--------------------------------------|
| GET    | <code>/api/guild/summary</code> | JWT               | Guild info + activity + achievements |
| GET    | <code>/api/guild/roster</code>  | guild.view_roster | Full roster with filters             |

## Roles & Permissions

| Method | Endpoint                              | Auth               | Description          |
|--------|---------------------------------------|--------------------|----------------------|
| GET    | <code>/api/roles</code>               | admin.manage_roles | List roles           |
| GET    | <code>/api/roles/permissions</code>   | admin.manage_roles | List all permissions |
| POST   | <code>/api/roles</code>               | admin.manage_roles | Create role          |
| PUT    | <code>/api/roles/:id</code>           | admin.manage_roles | Update role          |
| DELETE | <code>/api/roles/:id</code>           | admin.manage_roles | Delete role          |
| GET    | <code>/api/roles/users/:userId</code> | GM                 | Get user's roles     |
| PUT    | <code>/api/roles/users/:userId</code> | GM                 | Set user's roles     |



## Discord Roles

| Method | Endpoint                                    | Auth | Description         |
|--------|---------------------------------------------|------|---------------------|
| GET    | <code>/api/discord-roles/guild-roles</code> | GM   | Live Discord roles  |
| GET    | <code>/api/discord-roles/mappings</code>    | GM   | Saved role mappings |
| PUT    | <code>/api/discord-roles/mappings</code>    | GM   | Update all mappings |

## Upload & Carousel

| Method | Endpoint                                       | Auth    | Description                                      |
|--------|------------------------------------------------|---------|--------------------------------------------------|
| POST   | <code>/api/upload</code>                       | JWT     | Upload image (max 15MB, auto-compressed to WebP) |
| GET    | <code>/api/carousel</code>                     | —       | Carousel images + background                     |
| POST   | <code>/api/carousel</code>                     | Officer | Add carousel image                               |
| PUT    | <code>/api/carousel/:id</code>                 | Officer | Update carousel image                            |
| DELETE | <code>/api/carousel/:id</code>                 | Officer | Delete carousel image                            |
| GET    | <code>/api/carousel/settings</code>            | Officer | Get background image                             |
| PUT    | <code>/api/carousel/settings/background</code> | Officer | Set background image                             |

## Reports

| Method | Endpoint                                    | Auth             | Description             |
|--------|---------------------------------------------|------------------|-------------------------|
| GET    | <code>/api/reports/users</code>             | admin.view_panel | Member activity report  |
| GET    | <code>/api/reports/guild-gaps</code>        | admin.view_panel | Guild vs Discord gaps   |
| GET    | <code>/api/reports?status=open</code>       | admin.view_panel | Forum violation reports |
| PUT    | <code>/api/reports/:id</code>               | admin.view_panel | Update report status    |
| GET    | <code>/api/reports/users/presets</code>     | admin.view_panel | Saved report presets    |
| POST   | <code>/api/reports/users/presets</code>     | admin.view_panel | Save report preset      |
| DELETE | <code>/api/reports/users/presets/:id</code> | admin.view_panel | Delete preset           |

## Addon

| Method | Endpoint                     | Auth          | Description                           |
|--------|------------------------------|---------------|---------------------------------------|
| POST   | <code>/api/addon/sync</code> | JWT + Officer | Sync WoW addon data (schema v3 or v4) |

## Reconciliation

| Method | Endpoint                                                  | Auth          | Description                                                              |
|--------|-----------------------------------------------------------|---------------|--------------------------------------------------------------------------|
| POST   | <code>/api/reconciliation/addon-paste</code>              | JWT + Officer | Paste-based addon JSON ingest (backfill officer/public notes + lastSeen) |
| POST   | <code>/api/reconciliation/refresh</code>                  | JWT + Officer | Re-run Discord member sync + guild roster sync                           |
| POST   | <code>/api/reconciliation/guild-members/:id/link</code>   | JWT + Officer | Manually link a guild row to a site user                                 |
| POST   | <code>/api/reconciliation/guild-members/:id/ignore</code> | JWT + Officer | Mark guild row as intentionally unlinked                                 |
| DELETE | <code>/api/reconciliation/guild-members/:id/ignore</code> | JWT + Officer | Clear the ignore flag                                                    |

## 26. Troubleshooting

### Common Issues

#### "COLLATION is not valid for CHARACTER SET 'binary'"

Run the charset migration:

```
mysql -u USER -p DB_NAME < db/migration-040-fix-charset-utf8mb4.sql
```

#### Unicode characters not displaying (guild roster, leaderboards)

Same fix — migration-040 converts all tables to `utf8mb4`.

#### sharp fails to install on Linux

sharp requires native dependencies. On Ubuntu/Debian:

```
sudo apt-get install -y build-essential libvips-dev  
npm install sharp
```

#### PM2 not starting on boot

```
pm2 startup systemd  
# Run the sudo command it prints  
pm2 save
```

#### SyntaxError: Unexpected token '.' on startup

This means pm2 is running the app under an older Node (e.g. system `/usr/bin/node` which is Node 12 on the current production box) that can't parse optional chaining. The `interpreter` key in `ecosystem.config.js` pins the correct Node binary. If you see this error:

```
# Confirm which Node pm2 is actually using for the app
pm2 env 0 | grep node_version

# Fix: ensure ecosystem.config.js has interpreter set, then re-create the process
cd ~/MDGA && git pull
pm2 delete mdga && pm2 start ecosystem.config.js && pm2 save
```

### Frontend changes not showing

Rebuild the React app:

```
cd client && npm run build
```

### 5xx errors on character lookup

Check that `BLIZZARD_CLIENT_ID` and `BLIZZARD_CLIENT_SECRET` are set correctly in `.env`. The Blizzard token may have expired — restart the server to refresh it.

### Discord login redirect loop

Verify `DISCORD_REDIRECT_URI` in `.env` matches exactly what's configured in the Discord Developer Portal (including protocol, domain, and path).

### Images not loading after upload

Make sure `uploads/` directory exists and has write permissions:

```
mkdir -p uploads
chmod 755 uploads
```

### Database connection refused

Check MySQL is running and credentials in `.env` are correct:

```
mysql -h$DB_HOST -P$DB_PORT -u$DB_USER -p$DB_PASSWORD $DB_NAME -e "SELECT 1"
```